

Doxiveex 500mg/g Water Soluble Powder

Doxycycline hyclate.....100gm in 200gm

Product Description

It is a yellowish crystalline powder.

Pharmacodynamics Properties

Doxycycline belongs to the group of the tetracycline antibiotics. These antibiotics have a broad spectrum of antimicrobial activity, sharing the same basic structure of polycyclic naphthacenecarboxamide.

Doxycycline is primarily a bacteriostatic drug. It exerts its action by inhibiting the protein synthesis of the bacterial cell. Inhibition of bacterial protein synthesis results in disturbance of all functions necessary for the life of bacteria. Especially cell-division and the formation of the cell wall are impaired.

Doxycycline is a broad-spectrum antibiotic, active against a large number of Gram-positive and Gram-negative, aerobe and anaerobe micro-organisms, Mycoplasmata, Chlamydiae and Rickettsia.

For *Ornithobacterium rhinotracheale* results demonstrate a great variation from high to low susceptibility, depending on the geographical region where isolates came from.

Four resistance mechanisms acquired by microorganisms against tetracyclines in general have been reported: Decreased accumulation of tetracyclines (decreased permeability of the bacterial cell wall and active efflux), protein protection of the bacterial ribosome, enzymatic inactivation of the antibiotic and rRNA mutations (preventing the tetracycline binding to ribosome). Tetracycline resistance is usually acquired by means of plasmids or other mobile elements (e.g. conjugative transposones). Cross resistance between tetracyclines has also been described. Due to the greater liposolubility and greater facility to pass through cell membranes (in comparison to tetracycline), doxycycline retains a certain degree of efficacy against microorganisms with acquired resistance to tetracyclines.

Pharmacokinetics Properties

Doxycycline is absorbed in the stomach and the first part of the duodenum. Compared to the older tetracyclines the absorption of doxycycline is less affected by the presence of bivalent cations in food.

Following administration of doxycycline hyclate at an actual dose of 21 mg/kg body weight to chickens mean plasma concentrations above 1 µg/ml were reached within 6 hours and lasted for 6 hours after cessation of medication. From 24 h up to 96 h after start of treatment the doxycycline plasma concentrations exceeded 2 µg/ml. Following administration of doxycycline hyclate at an actual dose of 10 mg/kg body weight steady state plasma concentrations ranged from 0.75 to 0.93 µg/g between 12 and 96 hours after start of medication.

Because doxycycline is highly lipid soluble, it has a good tissue penetration. Respiratory tract tissue: plasma ratios of 1.3 (healthy lungs), 1.9 (pneumonic lungs) and 2.3 (nasal mucosa) have been reported for doxycycline. Plasma protein binding is high (over 90%).

Doxycycline is scarcely metabolised. Doxycycline is primarily excreted with the faeces.

Indication

Broilers, Breeders

Where clinical disease is present in the flock, to reduce mortality, morbidity, and clinical signs and to reduce lesions due to Pasteurellosis caused by *Pasteurella multocida* or to reduce morbidity and lesions in respiratory infections caused by *Ornithobacterium rhinotracheale* (ORT).

Dosage

The recommended dose is: 10 mg doxycycline hyclate (20 mg product) per kg body weight per day for 3-4 consecutive days in case of infections caused by *P. multocida* and 20 mg doxycycline hyclate (40 mg product) per kg body weight per day for 3-4 consecutive days in case of infections caused by *O. rhinotracheale*.

Mode of Administration

Administration orally with the drinking water.

Contraindication

Do not use in case of hypersensitivity to the active substance or to any of excipients.

Do not use in animals with an impaired liver function.

Warning and Precautions

Special precautions for use in animals

Due to likely variability (time, geographical) in susceptibility of bacteria for doxycycline, especially susceptibility of *O. rhinotracheale* may differ from country to country and even farm to farm, bacteriological sampling and susceptibility testing are recommended.

Use of the product should be based on culture and sensitivity of micro-organisms from diseased cases on farm. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria.

As eradication of the target pathogens may not be achieved, medication should therefore be combined with good management practices, e.g. good hygiene, proper ventilation, no overstocking.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

If you know you are allergic to the tetracycline class of antibiotics, special care should be taken when handling this product or the medicated solution.

During preparation and administration of the medicated drinking water, skin contact with the product and inhalation of dust particles should be avoided. Wear impermeable gloves (e.g. rubber or latex) and an appropriate dust mask (e.g. disposable half-mask respirator conforming to European Standard EN149) when applying the product.

In the event of eye or skin contact, rinse the affected area with large amounts of clean water and if irritation occurs, seek medical attention. Wash hands and contaminated skin immediately after handling the product.

If you develop symptoms following exposure such as skin rash, you should seek medical advice and show this warning to the physician. Swelling of the face, lips or eyes, or difficulty with breathing are more serious symptoms and require urgent medical attention.

Do not smoke, eat or drink while handling the product.

Take measures to avoid producing dust when incorporating the product into water. Avoid direct contact with skin and eyes when handling the product to prevent sensitisation and contact dermatitis.

Interaction with other medicine

Do not combine with antibiotics that are bacteriocidal e.g. penicillins or cephalosporins.

Absorption of doxycycline can be decreased in the presence of high quantities of calcium, iron, magnesium or aluminium in the diet. Do not administered together with antacids, kaolin and iron preparations.

It is advised that the interval between the administration of other products containing polyvalent cations should be 1-2 hours because they limit the absorption of tetracyclines.

Doxycycline increases the action of anticoagulants.

The solubility of the product is pH dependent and will precipitate if mixed in alkaline solution.

Do not store the drinking water in metallic containers.

Pregnancy and Lactation

Doxycycline has a low affinity for forming complexes with calcium and studies have demonstrated that doxycycline scarcely affects skeleton formation. No negative effects were observed in poultry after the administration of therapeutic doses of doxycycline.

Sides Effects

Tetracyclines may - in very rare cases - induce photosensitivity and allergic reactions. If suspected adverse reactions occur, treatment should be discontinued. Inform your veterinary surgeon if adverse reactions occur that are not stated.

Symptoms and treatment of overdose

Poultry tolerate double overdoses of doxycycline (40 mg/kg body weight) without any clinical effect.

Storage condition

store in a cool dry place below 30°C

Proposed shelf life

3 years

Packing

200gm

Manufacturer

Nam Pharma Sdn. Bhd.

5, Lebuhr Perusahaan Klebang 11, Taman Perindustrian Antarabangsa IGB, 31200 Chemor, Perak

Distributor

F.E Venture Sdn Bhd

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